

Review

# Metabolic mechanisms of species-specific developmental tempo

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## SUMMARY

Development consists of a highly ordered suite of steps and transitions, like choreography. Although these sequences are often evolutionarily conserved, they can display species variations in duration and speed, thereby modifying final organ size or function. Despite their evolutionary significance, the mechanisms underlying species-specific scaling of developmental tempo have remained unclear. Here, we will review recent findings that implicate global cellular mechanisms, particularly intermediary and protein metabolism, as species-specific modifiers of developmental tempo. In various systems, from somitic cell oscillations to neuronal development, metabolic pathways display species differences. These have been linked to mitochondrial metabolism, which can influence the species-specific speed of developmental transitions. Thus, intermediary metabolic pathways regulate developmental tempo together with other global processes, including proteostasis and chromatin remodeling. By linking metabolism and the evolution of developmental trajectories, these findings provide opportunities to decipher how species-specific cellular timing can influence organism fitness.

## TIME AS A LINKER OF DEVELOPMENT AND EVOLUTION

Developmental trajectories are often conserved throughout evolution, but they can display significant temporal differences depending on the species considered. Species-specific features in the timing of developmental events can be distinguished in various ways. In some cases, which can be referred to as heterochrony, there are species-specific changes in the relative order of several steps or in the duration of some steps relative to others within the same developmental sequence. In other cases, which can be referred to as allochry, the duration of an entire developmental sequence is scaled up or down, while the relative duration and order of each step are preserved.

Both heterochrony and allochry can profoundly influence the shape, size, and function of cells, organs, and organisms, thereby affecting fitness.<sup>1</sup> For instance, the sequential development of somites is orchestrated by somitic cell oscillations together with a wavefront of signaling pathways along the body axis,<sup>2</sup> which can display temporal variations across species, thereby influencing the size and number of somites. Similarly, the relative timing of gene expression patterns in the developing body and limbs displays species differences that influence the relative positioning of the limbs among tetrapods.<sup>3–5</sup> In other cases, the developmental sequence appears to be globally conserved but displays species-specific speed or duration. This is particularly important during human brain development, more specifically in the developing cerebral cortex.<sup>6</sup> During fetal stages, while cortical neurogenesis takes about 1 week in the mouse, it takes up to 4 months in the human, which contributes to brain size expansion by allowing increased rounds of self-renewing divisions and neuronal generation.<sup>7–10</sup> Postnatally,

cortical neurons undergo massive cellular and molecular changes to grow and refine synaptic connections and form neural circuits.<sup>11</sup> This maturation process is considerably prolonged in the human cortex. It takes a couple of weeks in mice or macaques, but over several years in the human cortex.<sup>6</sup> The resulting brain neoteny, or retention of juvenile brain features in a more mature organism, is thought to be required to optimize the functionality of neural circuits, for instance, by allowing prolonged periods of plasticity.<sup>12</sup>

## INTRINSIC CONTROL OF SPECIES-SPECIFIC DEVELOPMENTAL TEMPO: INSIGHTS FROM PLURIPOTENT STEM CELL MODELS

The examples above illustrate the evolutionary significance of heterochrony and allochry. But the mechanisms underlying species variations in developmental timing have remained largely unknown. This is partly due to the difficulties in studying these phenomena in the context of whole organisms. Every developmental step tends to depend on previous ones, and this inherent contingency makes it difficult to interpret timing differences and to untangle the underlying mechanisms.

Species-specific developmental timing can rely on extrinsic signals that coordinate transitions at the whole organism or organ level, like metamorphosis in some amphibians and insects, or pregnancy, delivery, and adolescence in mammals. However, recent evidence points to cell-intrinsic mechanisms acting as internal molecular hourglasses at the level of individual cells to set the species-specific tempo of developmental sequences. To reveal such mechanisms, the advent of pluripotent stem cell (PSC) technologies to model embryonic



development has proved to be particularly useful. Thanks to the availability of PSCs from an increased number of species and constantly improved models that mimic faithfully enough various key steps of development, it has become possible to compare developmental events of different species under similar conditions. In most systems, developmental timing has stood out among robust and substantial differences noted between PSC-derived development in various species, despite strong cellular and molecular conservation of most other aspects.<sup>13,14</sup>

For instance, *in vitro* models of cerebral cortex development revealed that the species-specific length of neurogenesis is retained *in vitro*, even when cortical cells are mixed with cells of other species.<sup>15–17</sup> Similarly, cortical neurons derived from human and non-human PSCs retain their species-specific speed of maturation *in vitro* or even following xenotransplantation.<sup>15–19</sup> Other pathways of neural development have been examined using PSCs from different species, leading to the same conclusion of large conservation of developmental tempo. For instance, mouse and human motor neurons were shown to display the same difference in timing *in vivo* and *in vitro*.<sup>20</sup> Importantly, however, the timing difference noted was about 2- to 3-fold, thus much less than the scale of timing difference (>10-fold) between human and non-human cortical cells. This indicates that cell-intrinsic differences in timing can be observed for similar orthologous cell types across species, but also for different cell types in the same species.

The *in vitro* retention of species-specific timing was observed for several other developmental trajectories, such as the differentiation of cardiomyocytes or pancreatic cells,<sup>21,22</sup> but also for oscillatory developmental processes, in particular the somite segmentation clock.<sup>23–25</sup> This clock controls the pace of sequential somite formation through a highly conserved oscillatory pattern of gene expression in presomitic mesodermal (PSM) cells.<sup>26</sup> Despite the highly conserved process of somitogenesis per se, the timing of somite formation and the period of oscillations of the clock vary substantially across vertebrates, from 30 min in the zebrafish to 2 h in the mouse and 4–5 h in the human species.<sup>27</sup> Changes in the segmentation clock can affect somite production: in snakes, the high number of somites in the thoracic and caudal regions is linked to an accelerated segmentation clock relative to axis elongation.<sup>28</sup> *In vitro* models using various species of PSCs have revealed that PSM cells retain the same species-specific oscillations as *in vivo*, again indicative of cell-intrinsic regulation.<sup>23–25</sup>

### SPECIFIC AND GLOBAL MECHANISMS TO SCALE DEVELOPMENTAL TIMING

These findings have revealed a robust cell-intrinsic component of the developmental timing differences across species, but what could be the underlying molecular mechanisms?

The usual suspects in evolutionary changes are expected to be found within the gene regulatory networks (GRNs) that control any given developmental trajectory. GRN modifications at the level of transcription factors, *cis*-regulatory elements, key effectors, or, most often, a combination thereof, can in principle explain most of the evolutionary changes in development<sup>29</sup> (Figures 1A and 1B). For instance, the relative timing of expres-

sion of Hox genes and of extrinsic cues (growth differentiation factor [GDF11] and retinoic acid) varies between vertebrate species, which impacts the differential positioning of the limbs.<sup>3–5</sup> Similarly, the prolonged expression of the Oct4 transcription factor during body axis extension in snakes, associated with specific *cis*-element rearrangements, has been linked to the elongation of their trunk.<sup>30</sup> The addition of species-specific genes to the ancestral GRN can also lead to temporal changes (Figure 1C). SRGAP2B/C human-specific genes are required for human cortical neuron neoteny by slowing down the maturation of their synapses.<sup>31,32</sup> Similarly, the NOTCH2NL genes are a group of hominid-specific genes that encode partial duplicates of the NOTCH2 gene.<sup>33–35</sup> By enhancing Notch signaling in human cortical progenitors, they contribute to a prolonged period of neurogenesis.<sup>33,34</sup>

These findings illustrate how species-specific gene expression temporal shifts and/or modification of discrete components of the GRN can influence the outcome of developmental trajectories and thereby shape organisms differently.

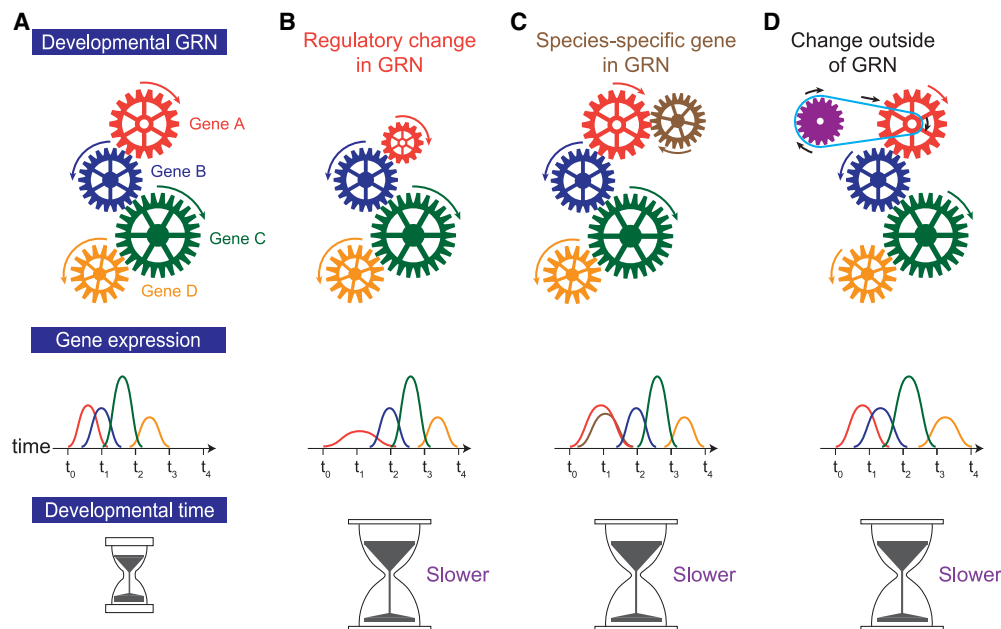
However, in many other developmental processes, species-specific changes in developmental timing may be observed in the absence of any detectable divergence in the underlying GRN. In other words, the sequence of events appears to be the same across species, but its global timescale has changed (Figure 1D). This was recently explored in two systems: the oscillations of the somitic clock and the genesis of spinal cord motor neurons.<sup>20,24</sup> In each case, a similar difference in timescale was observed between the two species, with mouse cells displaying a 2- to 3-fold faster tempo compared with human cells. However, an array of experimental approaches failed to reveal any significant divergence in the underlying GRNs (transcription factors or *cis*-regulatory elements) that could explain the differential tempo. Instead, species differences were found at the level of RNA and protein turnover rates, which were 2-fold higher in mouse than human cells, thus scaling with the differences in developmental timing.<sup>20,24</sup> Moreover, these correlations were recently extended to other mammalian species for PSM cells.<sup>25</sup>

These findings lead to the prediction that changes in rates of RNA and/or of protein turnover (including synthesis and/or degradation processes) could influence developmental tempo across species. However, this has remained difficult to test experimentally because of the inherent difficulties in manipulating the global rates of RNA/protein turnover with the required sensitivity and specificity.

### INTERMEDIARY METABOLISM: AN OLD ACTOR COMING BACK TO FAME

Studies in several systems have also shown that developmental timing can be directly influenced by “intermediary” metabolism,<sup>36,37</sup> which integrates all of the biochemical reactions that provide cells with energy, redox balance, and substrates for biosynthetic pathways on a global scale.

The notion that metabolic rates have an influence on species-specific properties is far from new. Species differences in metabolic rates have long been known to be anti-correlated with body mass, according to an allometric scaling relationship known as Kleiber’s law.<sup>38,39</sup> Importantly, the scaling factor that links metabolic rates and body size can vary, depending on both intrinsic



**Figure 1. Gene regulatory networks and developmental tempo**

(A) Gene regulatory networks (GRNs) comprise the elements controlling gene expression, their upstream regulators, and downstream effectors. In the context of species-specific developmental timing, they can be compared with the gears of a clock device that lead to the proper development of gene expression over time. Intrinsic and extrinsic factors can act as inputs that cause the gears to rotate, leading to a chain reaction of gene expression according to a specific timeline. The difference in size between the hourglasses illustrates variations in the speed and duration of development.

(B) Divergence in a specific component of the GRN (illustrated here by a change in the size of a specific [red] gear) can lead to local temporal changes, which, in turn, can lead to heterochrony (in this case, a delay).

(C) Similar effects can be induced by a more drastic change in the GRN, such as the inclusion of an additional species-specific gene (brown gear) that in this case acts as a brake to the function of an upstream component of the GRN.

(D) Global processes “outside” of the developmental GRN have an impact on timing without alteration of GRN components.

See text for more details.

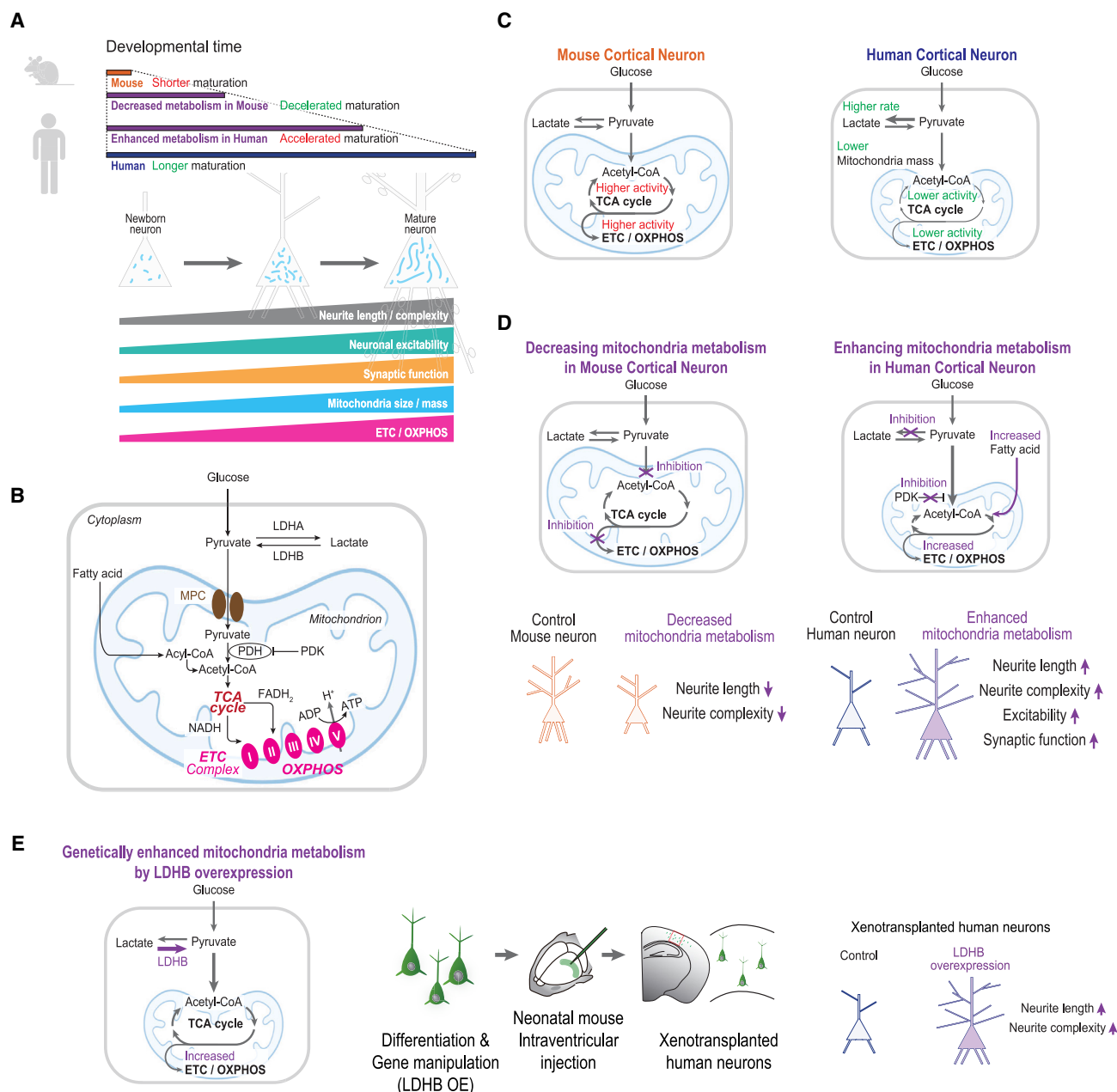
and extrinsic influences.<sup>40,41</sup> This suggests that the regulation of metabolic scaling is plastic and context-dependent. Metabolic rates have also been linked to lifespan across species: faster metabolic rates tend to be correlated with accelerated aging processes, resulting in shorter lifespans, although not all evidence points toward this conclusion.<sup>42,43</sup> On the other hand, genetic studies have demonstrated a causal relationship between lower mitochondrial metabolic rates and increased lifespan in several species.<sup>44–46</sup>

These studies suggest important links between species-specific metabolic rates, body growth, and aging, but molecular connections between metabolism and developmental trajectories have started to be unraveled more recently.<sup>47,48</sup> In particular, intermediary metabolism has been found to generate metabolites that can act as signaling or modulating factors for various molecular processes, particularly in stem cell fate transitions and embryogenesis.<sup>48–51</sup> Mitochondria are at the core of the regulation and generation of many of these metabolites, and, as such, these organelles are increasingly recognized as key players in developmental and stem cell biology.<sup>52,53</sup> Could mitochondrial metabolism display species differences at the cellular level and thereby contribute to species-specific developmental features? Two recent studies have tested this hypothesis and found that several components of intermediary metabolism display species differences and can influence developmental timing accordingly.<sup>36,37</sup>

## LINKING MITOCHONDRIAL METABOLISM WITH NEURONAL DEVELOPMENTAL TIMING

One study was focused on species-specific developmental timing of cortical neurons<sup>36</sup> in relation to human brain neoteny discussed above. Mitochondria have been causally linked to key aspects of neural development, from neurogenesis<sup>54–56</sup> to neuronal maturation and plasticity.<sup>56–59</sup> But whether and how this could be relevant to species differences in developmental timing had not been explored.

During cortical neurogenesis, neural progenitors generate neurons asynchronously, so that neurons with varying maturation states and identities coexist at each embryonic stage. This makes neuronal development particularly difficult to study with accurate temporal and cellular resolution. To circumvent this challenge, a genetic birth-dating system was developed to label cohorts of neurons born at the same time. Combining this approach with mitochondrial labeling in mouse and human cortical neurons *in vitro* revealed surprisingly dynamic changes over time and between species (Figure 2A). In newly generated cortical neurons, mitochondria were small in size and relatively sparse in both species. Mitochondria then gradually grew in number and size to reach maximal levels after 2–3 weeks in mouse cortical neurons (whether derived from *ex vivo* primary culture or PSCs), while it took several months in human cortical neurons (PSC-derived) to reach the same number and size of



**Figure 2. Species-specific mitochondrial metabolic activity regulates neuronal developmental tempo**

(A) Developing neurons of the cerebral cortex display species-specific temporal differences in mitochondrial growth and development, which are correlated with rates of neuronal maturation.

(B) A primer on intermediary metabolism. The glycolytic pathway converts glucose to pyruvate and produces ATP and nicotinamide adenine dinucleotide (NADH) in the cytoplasm. When pyruvate is produced, it undergoes conversion to lactate by two types of lactate dehydrogenase: LDHA and LDHB, which preferentially catalyze one of the two conversions. Cytoplasmic pyruvate can also enter mitochondria through a specific pyruvate carrier (MPC) and be converted to acetyl-CoA by pyruvate dehydrogenase complex (PDH) to feed the tricarboxylic acid (TCA) cycle, which generates NADH and flavin adenine dinucleotide (FADH<sub>2</sub>). Mitochondrial fatty acid  $\beta$ -oxidation (FAO) also produces acetyl-CoA, which enters the TCA cycle. The cofactors NADH and FADH<sub>2</sub> are used as fuel to generate ATP through the electron transport chain (ETC) during oxidative phosphorylation (OXPHOS).

(C) Species-specific mitochondrial development and metabolism. Mouse neurons (left) have greater mitochondrial mass and ETC/OXPHOS activity than human neurons (right), which display higher lactate production.

(D) Mitochondrial metabolism can be decreased in mouse cortical neurons (left) through inhibition of MPC or ETC complex, which slows down the developmental tempo, while mitochondrial metabolism can be enhanced in human cortical neurons (right) through inhibition of LDHA or PDK or addition of free fatty acid, leading to acceleration of developmental tempo.

(E) Xenotransplantation experiments allow testing of the impact of mitochondrial metabolism on the speed of neuronal development *in vivo*. Overexpression of LDHB in xenotransplanted neurons results in an increase of ETC/OXPHOS activity, which, in turn, accelerates neuronal morphogenesis.

mitochondria. Thus, mitochondria display a species-specific pattern of dynamics that correlates with neuronal maturation (Figure 2A). This species-specific timing was confirmed *in vivo* using transmission electron microscopy analyses, comparing mouse cortical neurons and human cortical neurons xenotransplanted in the mouse brain.

Metabolic function of mitochondria was also assessed during neuronal development by performing measurements of the mitochondrial electron transport chain (ETC) oxidative activity on cohorts of neurons born at distinct time points.<sup>36</sup> This revealed a gradual increase in both basal and stimulated mitochondrial respiration activity during neuronal development. Mouse neurons displayed a faster increase in mitochondrial activity and much higher rates of respiration activity compared with their human counterparts (Figures 2A–2C). Mouse neurons displayed higher levels of activity of specific ETC components (in particular complex IV) compared with human counterparts even at early stages, suggesting selective species-specific differences in metabolic oxidative activity, independently of mitochondrial development. Glucose metabolic tracing was performed to compare in more detail the metabolic profile of human and mouse developing cortical neurons at a similar age following their neurogenic birth date. This revealed an increase in the conversion rate of pyruvate into lactate in human cortical neurons, which takes place in the cytosol, together with decreased conversion of pyruvate into metabolites of the tricarboxylic acid (TCA) cycle, which takes place in mitochondria (Figures 2B and 2C).

Thus, human cortical neurons at early stages of their development display a metabolic profile highly reminiscent of “Warburg” aerobic glycolysis, usually found in proliferative cells.<sup>60</sup> Consistent with these findings, *in vivo* analyses of brain metabolic rates for oxygen and glucose consumption have demonstrated that the human cortex displays a prolonged pattern of aerobic glycolysis over several postnatal years, in particular in the prefrontal cortex.<sup>61</sup> This prolonged postnatal dependence on aerobic glycolysis in human neurons may also be consistent with the observation that the peak of brain glucose demand occurs during childhood in humans, whereas in macaque monkeys the peak occurs at birth.<sup>62,63</sup>

### MITOCHONDRIAL METABOLISM AS A PACEMAKER OF CELLULAR MATURATION

These findings indicate that mitochondrial metabolic activity and neuronal developmental tempo are strongly correlated, but this does not necessarily mean causality. Mitochondrial metabolic pathways were manipulated pharmacologically and genetically, followed by analyses of neuronal developmental rates. This was achieved in several ways, targeting enzymes involved in the metabolic pathways that display differences between mouse and human developing cortical neurons (Figure 2D). Human cortical neurons were treated with chemical agents, leading to inhibition of lactate dehydrogenase A (LDHA), which converts pyruvate into lactate, or to enhancement of pyruvate dehydrogenase (through inhibition of its negative regulator pyruvate dehydrogenase kinase [PDK]), which converts pyruvate into Acetyl-CoA that feeds into the TCA cycle (Figures 2B and 2C). Stimulation of mitochondria-dependent TCA was also obtained by adding free fatty acids (FFAs) to the neurons.

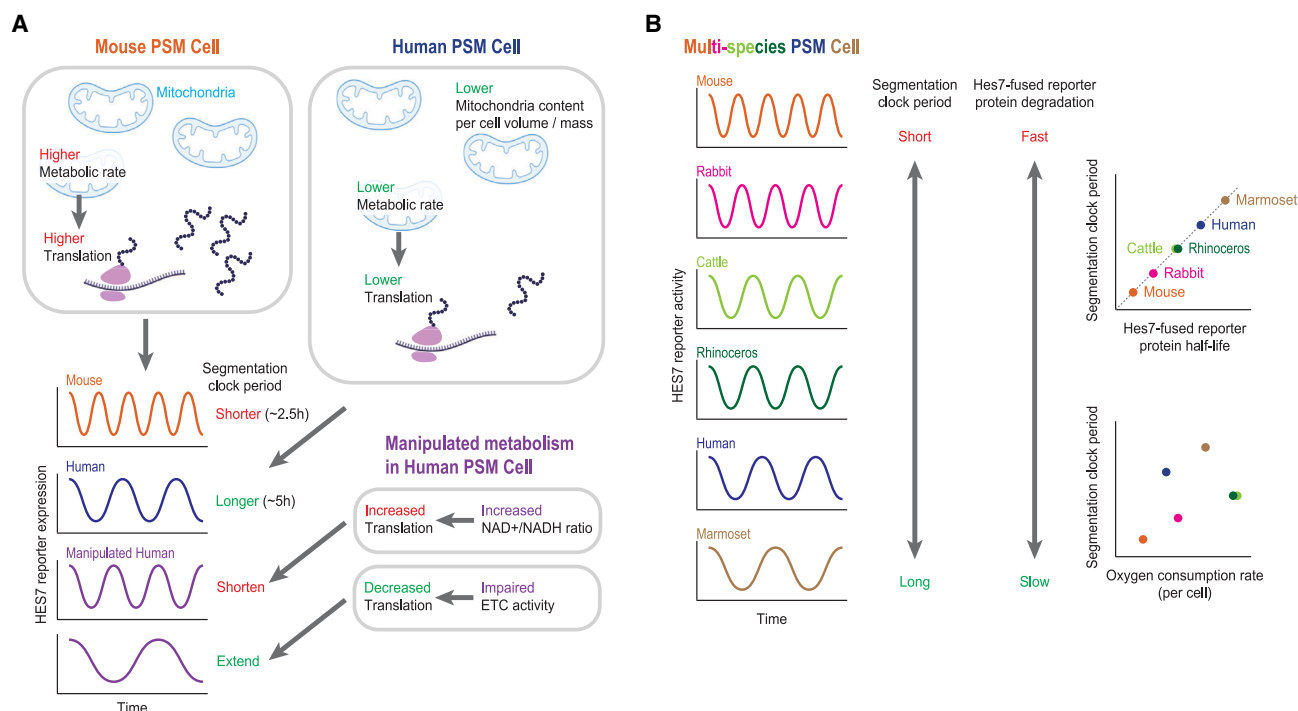
As expected, all these treatments led to enhanced mitochondrial oxidative activity in human neurons. Importantly, the treatments were stopped for 1 to 5 days before measuring neuronal maturation, to distinguish the effects on neuronal development from the acute impact of mitochondria on neuronal function. Maturation rates were then determined by measuring all key parameters of neuronal development, including cell size, neurite growth and complexity, electrical excitability, and synapse number and activity. All treatments boosting mitochondrial activity led neurons to undergo faster development, leading to the acquisition of maturation milestones weeks ahead of time (Figure 2B). Importantly, similar results were obtained *in vivo* following xenotransplantation of human cortical neurons in the mouse brain. Inducible overexpression of LDHB in transplanted human cortical neurons led to enhanced mitochondrial oxidative activity, followed by accelerated morphological development of the transplanted neurons (Figure 2E).

Finally, the impact of metabolic manipulation was examined on mouse cortical neurons. Also, in this case, mouse cortical neuron development was accelerated following mitochondria-boosting treatments, while, conversely, partial inhibition of mitochondrial activity (using blockade of pyruvate entry into mitochondria or inhibition of ETC I) resulted in their slower development (Figure 2D).

These results thus indicate that the tempo of cortical neuronal development is coupled to species-specific mitochondrial development and metabolism.

### MITOCHONDRIAL METABOLISM AND THE SOMITIC SEGMENTATION CLOCK

Could similar mechanisms operate in other systems, or is this phenomenon unique to neuronal development? Important insights have come in parallel from a recent study focusing on a completely different system—the somite segmentation clock found in PSM cells.<sup>37</sup> As explained above, human and mouse PSC-derived PSM cells display species-specific timing of gene expression oscillations, typically assessed with reporters of expression of the *Hes7* protein, a major driver of the somitic clock.<sup>23,24</sup> Examination of metabolic activity revealed that mouse PSM cells consume twice as much oxygen and glucose as their human counterparts when adjusting for cell mass. This increased metabolic activity was linked to mitochondrial biomass, which was doubled in the mouse vs. human PSM cells. Although the pharmacological inhibition of mitochondrial ETC oxidative activity (targeting complex I) could slow down the oscillation period of human PSM cells, the inhibition of ATP production (targeting ATP synthase) had no noticeable effect. This suggests that the period of the segmentation clock may not depend on mitochondria-dependent ATP production but rather on the mitochondrial REDOX (reduction-oxidation) status, which is also dependent on ETC (Figure 3A), through the oxidation of nicotinamide adenine dinucleotides (NADHs converted to NAD<sup>+</sup>). This was tested by manipulating NAD<sup>+</sup> levels using the *Lactobacillus brevis* NADH oxidase (LbNOX) enzyme, a bacterial NADH oxidase that can increase NAD<sup>+</sup> levels in eukaryote cells.<sup>64</sup> Overexpression of LbNOX in human PSM cells resulted in a transient acceleration of the clock, consistent with the direct effect of NAD<sup>+</sup> levels on PSM developmental tempo (Figure 3A).



**Figure 3. Species-specific metabolic activity regulates developmental tempo of presomitic mesoderm cells**

(A) Mouse presomitic mesoderm (PSM) cells display higher rates of mitochondrial metabolism and translation and a faster segmentation clock compared with human PSM cells. Inhibition of ETC activity reduces the rate of translation, extending the period of the segmentation clock. Conversely, an increased  $NAD^+/NADH$  ratio increases the rate of translation, shortening the period of the segmentation clock.

(B) Comparison of PSM cells derived from multiple mammalian species reveals the linear correlations between the segmentation clock period and Hes7-fused reporter protein half-life. However, no such correlation is detectable between the segmentation clock period and oxygen consumption rates.

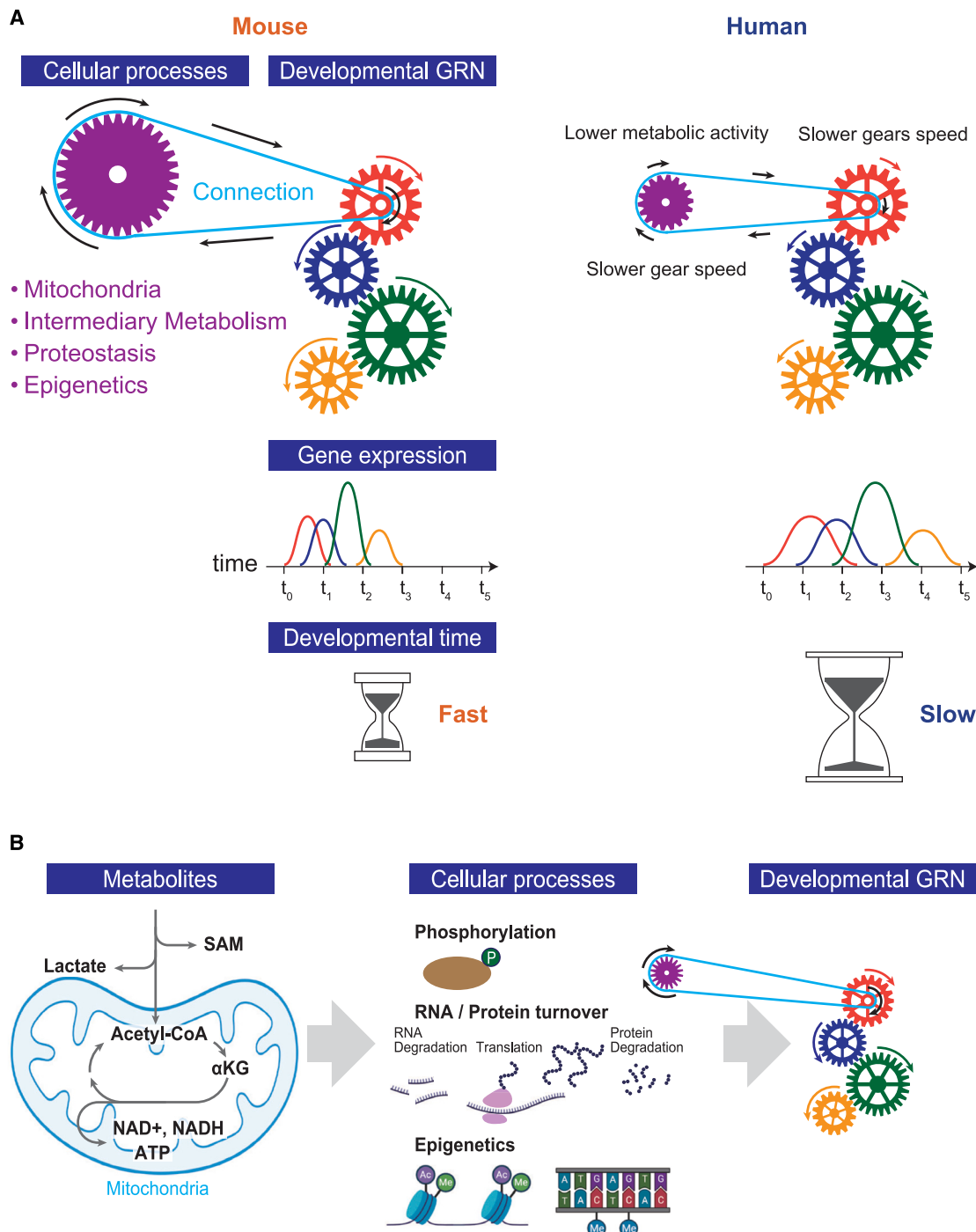
On the other hand, another recent study explored the relationships between the somitic clock and metabolic rates, focusing on global oxygen consumption rates but using PSM cells from 6 different species.<sup>25</sup> Although the species differences reported for human and mouse PSM cells could be confirmed, the species-specific somitic clock period was not correlated with oxygen consumption rate across all 6 species, although they could be linked with Hes7 protein turnover rates (Figure 3B).

Taken together, these findings indicate that mitochondrial metabolic activity can influence the PSM clock, but future work is needed to probe more deeply into its links with developmental timing across species. The exact contribution of the activity of each ETC component and the specific metabolites that may act as effectors of temporal regulation across species remain to be examined, such as ETC I that contributes the most to  $NAD^+/NADH$ , for instance, but also glycolysis and TCA metabolites. These cannot be resolved by global rates of oxygen consumption alone and will require more in-depth metabolomic measurements. Conversely, it will be interesting to test whether the links between species-specific mitochondrial metabolism and neuronal development found in mouse and human species can be extended to other species and cell types. Consistent with the latter, the maturation rates of cardiomyocytes and pancreatic beta cells can be enhanced by treatments that increase mitochondrial metabolism.<sup>65–67</sup> Finally, the links between intermediary metabolism and other global cellular processes that may act downstream or in parallel should be explored further, as discussed below.

## POTENTIAL DOWNSTREAM MECHANISMS LINKING METABOLISM AND DEVELOPMENTAL TIMING

Metabolic activity can thus act as an external force to drive GRN gears at a different speed (Figure 4A). But what could be the mechanisms linking metabolism and GRN temporal regulation? A first obvious target is the turnover of proteins, or proteostasis. Indeed, proteostasis varies across species,<sup>68</sup> its levels are correlated with developmental timing in several systems,<sup>20,25,69</sup> and it necessitates a substantial amount of energy and metabolites.<sup>70</sup> Indeed, an upsurge in mitochondrial mass or membrane potential results in a corresponding increase in the rate of transcription and translation per unit volume,<sup>71</sup> and mitochondrial activity contributes to up to 50% of the cellular variation in translation rate in a given cell type.<sup>71</sup>

A possible link between intermediary metabolism and protein turnover was explored in the PSM model.<sup>37</sup> Protein synthesis rates were found to be higher in mouse than human PSM, in correlation with the somitic clock differences. Pharmacological inhibition of mitochondrial ETC activity in human PSM reduced protein translation rates, together with a reduction in the clock oscillation period. Conversely, an increase in  $NAD^+/NADH$  levels induced by LbNOX led to an increase in the rates of protein synthesis, while protein synthesis inhibition could decrease the period of Hes7 oscillations in human PSM cells.<sup>37</sup> On the other hand, while proteasome activity was also found to display higher levels in mouse than human PSM cells, its inhibition in human PSM cells did not lead to major changes in the clock period.<sup>37</sup>



**Figure 4. Potential models and mechanisms linking metabolism and developmental time**

(A) External forces outside of the GRN can regulate its speed. Mitochondrial metabolic rates, protein turnover, translation speed, and epigenetic modulation are some of the key external forces that may impact the rotational speed of the gears of the developmental GRN and thereby developmental time. The connection illustrates that the GRN also has the ability to regulate external forces on metabolic processes. The mouse is an example of a faster-developing organism, while the human is an example of a slower-developing organism.

(B) Mitochondrial metabolites act as key substrates or regulators of global processes, including RNA and protein turnover, protein post-translational modifications, and chromatin remodeling, which, in turn, impact GRN developmental speed.

These findings indicate that protein synthesis acts downstream of oxidative activity to regulate the somitic clock speed (Figure 3A). It will be fascinating to test whether the same applies

to other systems that operate according to different timescales, such as neurogenesis, which spans weeks to months in most mammals and is highly dependent on translational control.<sup>72</sup>

Intermediary metabolism could also influence developmental tempo through metabolite-dependent post-translational modifications,<sup>48</sup> in particular histone acetylation and methylation patterns (Figure 4B). Several mitochondria and cytosolic metabolites are essential substrates or regulators of the enzymes required to generate and modify histone epigenetic marks in the nucleus, including DNA methylation (S-adenosyl methionine for methylation and  $\alpha$ -ketoglutarate for demethylation) and acetylation (acetyl-CoA for acetylation and NAD<sup>+</sup>/NADH as regulators of Sirtuin deacetylases).<sup>48</sup> For instance, metabolic and transcriptional analyses have shown that PSCs depend on glycolysis to produce acetyl-CoA. Acetyl-CoA production helps maintain histone acetylation, which is required for the expression of pluripotency genes and the maintenance of pluripotency. Pharmacological inhibition of glycolysis leads to histone deacetylation and differentiation of the PSCs. Conversely, supplementation of acetate, a precursor of acetyl-CoA, delays differentiation and decreases levels of histone deacetylation.<sup>73</sup>

Moreover, epigenetic remodelers play a crucial role in most cell fate transitions, including neurogenesis and neuronal development. The differentiation of cortical progenitors into cortical neurons requires the mitochondria-dependent histone deacetylation by NAD<sup>+</sup> sensor sirtuins.<sup>74,75</sup> Neuronal maturation is also highly dependent on chromatin remodeling, and changes in histone methylation dynamics were recently found to impact the speed of neuronal development in human and mouse neurons.<sup>76–79</sup> During human cortical neuron development, an epigenetic barrier was identified, consisting mostly of histone methylation marks, which counteracts differentiation. The barrier can be disrupted already at the level of cortical progenitor cells by inhibiting histone methyltransferase activity, which leads later on to acceleration of cortical neuron maturation, both morphologically and functionally.<sup>79</sup> These findings highlight the influence of early cellular events on the timing of later-stage development.

Altogether, these data suggest that mitochondrial metabolites could influence developmental timing through the control of histone modification rates. However, the evidence for interspecies differences in relevant metabolites or rates of chromatin remodeling is still missing. Conversely, epigenetic modifications could impact the expression dynamics of mitochondria or metabolic genes, thereby affecting metabolic rates (Figure 4A). Such an interplay between metabolism and epigenetic remodeling could be important to maintain species-specific metabolic rates in different environments, as observed during xenotransplantation. Moreover, it should be noted that other proteins than histones could be regulated by metabolite-dependent post-translational modifications,<sup>80</sup> which could be relevant for developmental timing as well.

Finally, metabolism could also exert its influence through the control of protein phosphorylation (Figure 4B). For example, AMP-activated protein kinase (AMPK) acts as a metabolic sensor for the ATP-to-ADP or ATP-to-AMP ratio. When activated, it phosphorylates key substrates, including histone acetyltransferase p300 and mammalian Target of Rapamycin (mTOR) complex 1 proteins,<sup>81</sup> and can influence cellular morphogenesis.<sup>82</sup> It remains to be tested whether mTOR or its upstream regulators are indeed involved in developmental timing and whether metabolic rates may influence mTOR in a species-specific fashion. Intriguingly, mTOR signaling displays higher levels in human neural progenitors compared with those

of other species<sup>83</sup> and is regulated by human-specific genetic pathways.<sup>84</sup>

Although protein turnover and post-translational modifications appear to be the most attractive mechanisms operating downstream of metabolism to link with developmental timing, other mechanisms also remain to be explored, such as rates of RNA transcription and splicing, for instance. In any case, these mechanisms should not be thought of as mutually exclusive or acting only in hierarchical ways. They are more likely to act in concert to regulate tempo, as illustrated by the impact of translation control of epigenetic remodelers to modulate rates of neurogenesis.<sup>85</sup>

### MITOCHONDRIA AND UPSTREAM MECHANISMS OF METABOLIC SPECIES SPECIFICITY

Another unresolved question raised by the recent studies concerns the upstream mechanisms that may underlie species and cell-specific metabolic patterns. To address this issue, mitochondria appear to be a good place to start. The relative biomass of mitochondria was found to be higher in mouse than in human PSM,<sup>37</sup> which could contribute to higher ratios of NAD<sup>+</sup>/NADH. In developing neurons, mitochondria display species-specific patterns of mitochondrial dynamics and biogenesis that cause differences in neuronal speed.<sup>52</sup> In at least some of the species, differences in mitochondrial features are of transcriptional origin: many nuclear genes encoding mitochondrial proteins display dynamic temporal regulation during cortical neuron development at a species-specific pace,<sup>52</sup> suggesting that mitochondrial gene regulation is somehow coordinated with cell development.

Additionally, post-transcriptional levels of regulation may contribute to species-specific mitochondrial features. The mammalian mitochondrial proteome is composed of 1,000 to 1,500 proteins, with varying functions across different tissues and cell types, thanks to the diversity in their protein composition.<sup>86</sup> Novel tools enabling high-throughput, cell-specific proteomic analyses of mitochondria<sup>87</sup> should enable exploration of the properties of mitochondria across species and models, potentially linking them with developmental tempo. It will be particularly interesting to find out whether there are any specific mitochondrial components that can lead to species-specific functionality at whole organismal or cell-specific levels. For example, naked mole-rat mitochondria display lower levels of oxidative activity but also greater ability to scavenge reactive oxygen species compared with laboratory mice,<sup>88</sup> while human neural progenitors' mitochondria express a human-specific protein that modifies glutamine entry into the TCA cycle.<sup>89</sup>

### CONCLUSIONS, UNRESOLVED QUESTIONS, AND FUTURE DIRECTIONS

The mechanisms underlying species-specific developmental timing have long remained mysterious, but recent findings point to a number of mechanisms, which mostly relate to global cellular processes—in particular, intermediary metabolism and protein turnover. It is remarkable that similar observations have been made independently in very distinct systems, from somitic cells to cortical neurons. While this suggests that similar “universal” mechanisms may regulate developmental timing across many systems and species, their generality remains to be tested

in depth. Alternatively, specific metabolic or other processes may influence timing in various ways, depending on the cellular context or temporal scaling considered, from hours-long transitions that affect many different cell types in a coordinate fashion during early embryogenesis, to highly cell-specific changes leading to years-long effects on organism function.

Moreover, interactions between different cell types, tissues, and external factors that influence developmental transitions should be explored further in the context of species-specific timing. For instance, amyloid precursor protein (APP) was recently found to maintain prolonged neurogenesis in human cortical progenitors through facilitation of Wnt signaling, while APP's impact was negligible in mouse cell counterparts.<sup>90</sup> Thus, differential regulation of extrinsic cues could also contribute to developmental timing differences across species. A recent study reported that human PSCs appear to differentiate faster into neural cells when co-cultured with an excess amount of equivalent-stage mouse PSCs.<sup>91</sup> This suggests that extrinsic cues could potentially alter the species-specific intrinsic pace of PSC differentiation. However, their molecular identity remains to be identified, as well as their physiological contribution to developmental timing across species. To resolve these questions, further research should delve deeper into the underlying processes at the highest possible molecular and cellular resolution, using multiple systems and species.

Whatever the mechanisms involved in species-specific developmental timing, their better understanding provides promising experimental venues to explore the impact of developmental timing on organ and organism function and fitness.

Indeed, the recent findings described above provide new ways in which to manipulate developmental timing and examine the consequences on species-specific outcomes and fitness. For example, one could examine the long-term impact of temporal changes in the somitic clock or other early events of embryogenesis on final body size and shape. Similarly, molecular manipulations leading to prolonged neurogenesis could be used to modify brain size and composition, followed by functional and behavioral investigations.<sup>92</sup> These types of experiments could determine whether and how specific changes in developmental timing can be either mostly adaptive during development, for instance, to ensure more robustness through slower developmental transitions,<sup>93</sup> or lead to long-term consequences—or even emerging properties—in adult organism fitness. In this respect, it will be particularly interesting to determine the potential impact of highly derived features, such as human cortical neuron neoteny, on brain circuit function and dysfunction.<sup>6</sup> Conversely, the identification of the mechanisms responsible for developmental timing will enable the determination of their potential link with human diseases. For instance, intellectual disabilities and autism spectrum disorders have been proposed to be linked to dysregulation of brain developmental timing, which should be examined mechanistically.<sup>32,94–96</sup>

Finally, metabolic reprogramming could be used in principle to accelerate cell differentiation, which could greatly facilitate human stem cell modeling of mature cell function and diseases.

To conclude, many questions remain in respect of the mechanisms underlying species-specific developmental timing and their significance for evolution and diseases. But the recent findings described here provide a first important framework to

address them across systems and species. What will the answers to these fascinating questions be? Time will tell.

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## DECLARATION OF INTERESTS

The authors declare no competing interests.

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